Rui Xing

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EDUCATION BACKGROUND

PhD in Engineering (Computer Science) **2022—Present**

University of Melbourne Melbourne, Australia

M.Eng. in Software Engineering **2017—2020**

Beihang University (Beijing University of Aeronautics and Astronautics, BUAA) Beijing, China

B.Eng. in Computer Science and Technology **2013—2017**

HeFei University of Technology (HFUT) Anhui, China

PUBLICATIONS

- **Rui Xing**, Jie Luo, Tengwei Song. (2020) "BioRel: towards large-scale biomedical relation extraction." BMC bioinformatics 21.16 (2020): 1-13.
- **Rui Xing** and Jie Luo. (2019). Distant Supervised Relation Extraction with Separate Head-Tail CNN. In Proceedings of the 5th Workshop on Noisy User-generated Text (EMNLP W-NUT 2019), 249–258.

RESEARCH EXPERIENCE

Empirical Method of Biomedical Knowledge Graph Construction Apr. 2019—Aug. 2019

- Proposed a new large-scale dataset, *BioRel*, for distant supervised relation extraction in biomedical domain. The dataset is created by aligning UMLS relations to Metamapped Medline Corpus.
- Conducted comprehensive experiments to show that the dataset is suitable for the state-of-the art neural-based methods compared to traditional statistic methods. Further analysis also suggested the dataset have fewer noisy instances.

Distant Supervised Relation Extraction Sept. 2018—Mar. 2019

- Designed separate head-tail CNN to alleviate mislabeling issue in distant supervised relation extraction.
- Proposed separate head-tail convolution and pooling to extract better semantic features, and classified relation from coarse-to-fine to alleviate impact of Not-A relation instances.
- Implemented full model and conducted experiments to show the method increases precision by 8%.

Relation Extraction from Biomedical Text Dec. 2017—Jul. 2018

- Introduced bootstrapping framework to extract binary relation in biomedical text, which discovered relational facts iteratively using entity pairs as seeds.

- Modified confidence of generated patterns and tuples and improved accuracy by 2%.

TEACHING

Beihang University, Beijing, China

Cloud Computing Spring Course 2019

Software Engineering Summer School (Beihang University & Microsoft Research Asia) 2018

SELECTED HONOURS & AWARDS

China Scholarship Council - University of Melbourne PhD Scholarship 2020

Excellent Graduate Student, Beihang University 2020

First-Class Scholarship for Postgraduate Student, Beihang University 2017

Honorable Mention in Mathematical Contest in Modeling, Consortium for Mathematics and Its Application 2016

National Scholarship, Ministry of Education in China 2015, 2016

Second Prize in RoboCup Soccer China Open 2D Simulation 2015

PROFESSIONAL EXPERIENCE

Reading group organizer, NLP group, University of Melbourne 2022-Present

Big data Intelligence group, Beihang University 2017-2020

2D Simulation League member, RoboCup Lab, HeFei University of Technology 2014-2017

Student Union member, HeFei University of Technology 2014-2015

PERSONAL

Skills

- Natural languages: English, Mandarin
- Programming languages: Python, Bash, Latex

Interests

- Swimming, badminton and table tennis